

An Asymptotic Lower Bound for the First Negative Thue–Morse Pressure Degree

Research note prepared for Vladimir Reshetnikov with OpenAI

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Abstract

For every sufficiently large integer m , all even coefficients of integer Thue–Morse pressure at degrees $2m < N \leq 33m/5$ are strictly positive. Thus the first negative even degree N_m above the cancellation at $2m$ satisfies $\liminf_{m \rightarrow \infty} N_m/m \geq 6.6$. The proof extends the previously established all-order window below $6m$. A full-disk bound controls the quadratic Schur term beyond the range of its unregularized orbit formula, and a new logarithmic-squared envelope controls the cubic term. An exact rational endpoint certificate and a uniform tilted local limit theorem then give one common eventual cutoff. No numerical cutoff, limiting slope, or exact first-negative formula is asserted.

Status. This is unrefereed ordinary mathematics, not a Lean formalization. It uses earlier proofs included with the sources and classical local-limit methods. No priority claim for those methods is made.

1 Statement and normalization

In the cosine-phase normalization of the repository manuscript [1], write

$$P_m(t) = p_m(t/\pi), \quad d = 2m, \quad m \geq 2.$$

Let N_m be the least even integer $N > d$ for which $[t^N]P_m(t) < 0$. The prior infinite-sign theorem [3] ensures that this minimum exists. Lower negative coefficients and the cancellation at degree d are not included in the definition.

Theorem 1.1. *There is an integer m_0 such that, for every integer $m \geq m_0$ and every even integer N with*

$$2m < N \leq \frac{33m}{5},$$

one has $[t^N]P_m(t) > 0$. In particular,

$$\liminf_{m \rightarrow \infty} \frac{N_m}{m} \geq \frac{33}{5}.$$

The existing theorem [2] gives positivity for every $m \geq 2$ throughout $d < N < 3d$. We prove the additional interval $3d \leq N \leq 33d/10$ uniformly for all sufficiently large d .

Editorial note (ProveIt, 2026-10-01). The bound on $\liminf N_m/m$ is implied by Theorem 1.1 of `../Thue_Morse_Integer_Pressure_First_Negative/`, $N_m/m \rightarrow \gamma > 6.662966 > 33/5$, which uses this theorem, Lemma 2.1 and the Schur expansion (1) as inputs; the strict positivity for $2m < N \leq 33m/5$ and $m \geq m_0$ is not implied by that statement. The certified value $N_{50} = 330 = 33 \cdot 50/5$ of `../Thue_Morse_Integer_Pressure_First_Negative_Data/` shows that $m_0 \geq 51$.

The normalized transfer operator is

$$(V_a f)(x) = \sum_{2y=x \pmod{1}} (\cos \pi y + a \sin \pi y)^d f(y).$$

Its atomic fixed function h_0 is normalized by $h_0(0) = 1$. The simple eigenvalue $\lambda(a)$ near one and its normalized eigenfunction satisfy

$$P_m(t) = d \log \cos t + \log \lambda(\tan t), \quad \lambda(a) - 1 = a^d h_a(1/2).$$

Put $\text{ev } f = f(1/2)$, $R_m = \text{ev } h_0$, $Qf = f - f(0)h_0$, and, on the subspace vanishing at zero,

$$\mathcal{B}_a = QV_a Q, \quad q_a = QV_a h_0.$$

Define the even analytic germs

$$\begin{aligned} F &= R_m + \text{ev}(I - \mathcal{B}_a)^{-1} q_a, \\ G &= \text{ev}(I - \mathcal{B}_a)^{-2} q_a, \quad H = \text{ev}(I - \mathcal{B}_a)^{-3} q_a. \end{aligned}$$

Here $G(0) = H(0) = 0$. The scalar $H(a)$ is distinct from the product $\mathcal{H}_a(x)$ introduced below.

2 The exact third Schur term

Set $\delta = \lambda - 1$ and $z = a^d$. The projected eigenvalue equation gives

$$\delta = z\{F - \delta G + \delta^2 H + O(\delta^3)\}.$$

Substitution and then expansion of $\log(1 + \delta)$ yield

$$\begin{aligned} \delta &= zF - z^2 FG + z^3(FG^2 + F^2 H) + O(a^{4d}), \\ \log \lambda &= zF - z^2 K_2 + z^3 K_3 + O(a^{4d}), \\ K_2 &= FG + F^2/2, \quad K_3 = FG^2 + F^2 H + F^2 G + F^3/3. \end{aligned} \tag{1}$$

All dependence of F, G, H on a is retained. Thus, below degree $4d$,

$$P_m(t) = d \log \cos t + \tan^d t F(\tan t) - \tan^{2d} t K_2(\tan t) + \tan^{3d} t K_3(\tan t). \tag{2}$$

This is an identity of Taylor coefficients; it asserts no separate sign for the terms.

Put

$$c = \frac{2}{\pi}, \quad b_j = \tan \frac{\pi}{2^{j+1}}, \quad B(a) = \prod_{j \geq 1} (1 + ab_j), \quad g(a) = caB(a), \quad \mathcal{G}(t) = g(\tan t).$$

The product B is entire, and $\mathcal{G}$ has strictly positive coefficients at every degree at least one.

Lemma 2.1. *For every even $d \geq 4$ and every even N with $3d \leq N < 4d$,*

$$\left| [t^N] \tan^{3d} t K_3(\tan t) \right| \leq 220(\log_2 d + 4)^2 [t^N] \mathcal{G}(t)^{3d}.$$

In particular, throughout $3d \leq N \leq 7d/2$ this is less than $220(\log_2 d + 4)^2 2^d$.

Proof. Write $M_q = [a^q]B(a)^d$. The established tangent majorization and marked-depth orbit formula, restated and proved in Section 2 of [2], give

$$|[a^q]F| \leq \frac{5}{2} c^d M_q, \quad |[a^q]G| \leq (5 \log_2 d + 18) c^d M_q \quad (0 \leq q \leq d-2). \tag{3}$$

We give the extra marking needed for H . For positive odd n , let $x_j(n) = |\tan(n\pi/2^{j+1})|/n$. A degree- q monomial in the orbit product has weight one in F , its deepest occupied index L in G , and $\binom{L+1}{2}$ in H . Indeed, below degree d all positive-degree insertions vanish at zero, so the Q projections can be removed. In the expansion of $(I - \mathcal{B}_a)^{-3}q_a$, the depth ℓ has weight $\binom{\ell+2}{2}$, while a monomial first enters the finite iterate at $L = \ell + 1$.

Tangent majorization bounds the absolute unmarked degree- q sum by M_q . Marking one selected unit beyond J bounds the corresponding tail by

$$M_q \min(1, \pi d^2 2^{-J}), \quad J \geq \lceil \log_2 n \rceil.$$

This uses $M_{q-1}/M_q \leq q/(d-q+1)$ and the geometric tangent tail, exactly as in [2]. At $q = 0$, G and H vanish and are handled directly.

Set $K = \max(\lceil \log_2 n \rceil, \lceil \log_2(\pi d^2) \rceil)$. Since

$$\binom{L+1}{2} = \sum_{J \geq 0} (J+1) \mathbf{1}_{L > J},$$

the marked sum is at most

$$M_q \left\{ \frac{(K+1)(K+2)}{2} + K + 3 \right\} \leq M_q (\log_2 n + 2 \log_2 d + 7)^2.$$

The pair of half-integer branches $\pm n/2$ contributes $2c^d n^{q-d}$. The bounds

$$\sum_{n \geq 1 \text{ odd}} n^{q-d} \leq \frac{5}{4}, \quad \sum_{n \geq 1 \text{ odd}} n^{q-d} (\log_2 n)^2 \leq 11$$

follow from $q \leq d-2$; for the second, enlarge to all $n \geq 2$ and sum $\sum_{k \geq 1} (k+1)^2 2^{-k} = 11$. Thus

$$|[a^q]H| \leq \{5(2 \log_2 d + 7)^2 + 44\} c^d M_q. \quad (4)$$

These summable bounds justify the marked orbit interchanges as well.

If $A = \log_2 d + 4 \geq 6$, the constants in (3)–(4) are at most $5/2$, $5A$, and $24A^2$. Convolving in K_3 gives the constant

$$\left(\frac{125}{2} + 150 \right) A^2 + \frac{125}{4} A + \frac{125}{24} < 220A^2.$$

Positive tangent coefficients prove the first assertion. Finally, $\sum b_j < 2$ and

$$\tan(1/8) \leq 16/127 < 63/500, \quad e^{63/250} < 13/10$$

imply $\mathcal{G}(1/8) < 273/2500 < 1/9$. The exponential inequality follows from $e^x \leq 1 + x + x^2/[2(1 - x/3)]$ for $0 \leq x < 3$. Hence $[t^N] \mathcal{G}^{3d} < 9^{-3d} 8^N < 2^d$ for $N \leq 7d/2$, since $8^7 < 4 \cdot 9^6$. $\square$

3 The weighted Green inputs

The only operator estimates imported from [2], Sections 4–5, are recorded here explicitly. Their full proofs, scalar certificates, and source are included unchanged with this package. Set

$$R = \frac{2}{5}, \quad w = \frac{49}{100}, \quad \mathcal{H}_a(x) = \prod_{j \geq 1} \left(\cos \frac{\pi x}{2^j} + a \sin \frac{\pi x}{2^j} \right), \quad h = \mathcal{H}_R, \quad L = h(1/2).$$

For even $d \geq 128$ and $|a| \leq R$, on $X = \{f \in C^1(\mathbb{R}/\mathbb{Z}) : f(0) = 0\}$ define

$$N_0(f) = \sup \frac{|f(x)|}{t(x)h(t(x))^d}, \quad N_1(f) = \sup \frac{|f'(x)|}{h(t(x))^d}, \quad t(x) = \text{dist}(x, \mathbb{Z}),$$

with the first supremum over nonintegers. Put $\eta_d = (9/10)^d/d$ and $\|f\|_d = N_0(f) + \eta_d N_1(f)$. Then

$$\|(I - QV_a|_X)^{-1}\| \leq 5, \quad |\text{ev } f| \leq \frac{1}{2}L^d\|f\|_d. \quad (5)$$

The inverse is analytic on a neighborhood of the closed disk. Also

$$1 \leq h \leq 7/5, \quad L < 243/200, \quad RL < w < 1/2, \quad (6)$$

h increases on $[0, 1/4]$, and every individual inverse branch satisfies $|W_a(y)|h(t(y))/h(t(x)) \leq 1$, where $W_a(y) = \cos \pi y + a \sin \pi y$.

The normalized frozen function

$$f_F = h_0 + (I - QV_a|_X)^{-1}QV_a h_0$$

has the approximation $f_F = \phi - C + e$, where

$$\phi(x) = \mathcal{H}_a(x)^d + \mathcal{H}_a(x-1)^d,$$

C is the cubic Hermite correction of the excess endpoint values and derivatives, and

$$\|C\|_\infty, \|C'\|_\infty \leq 610d(RL)^d, \quad \|e\|_d \leq 60000d^2w^d. \quad (7)$$

The corrected function is periodic and normalized at zero. The scalar consequence is

$$|F(a) - c^d(B(a)^d + B(-a)^d)| \leq 31000d^2(wL)^d \leq 31000d^2\beta^d, \quad \beta = \frac{11907}{20000}. \quad (8)$$

These inputs retain the actual weighted residual; no additional exponential transfer loss is introduced.

4 A full quadratic bound

Lemma 4.1. *For even $d \geq 128$ and $|a| \leq R$,*

$$|G(a)| \leq 4275dL^d, \quad |K_2(a)| \leq 13000dL^{2d}.$$

Proof. For $|u| \leq 1$, factorwise absolute values give $|\mathcal{H}_a(u)| \leq \mathcal{H}_R(|u|)$. The individual branch inequality, at $a = -R$, gives $\mathcal{H}_R(1-t) \leq h(t)$ for $0 \leq t \leq 1/2$. Both terms in ϕ therefore have weighted magnitude at most one. The bound $|\mathcal{H}'_a| < 120$ on $[-2, 2]$, proved in [2], gives

$$\frac{|\phi|}{h(t)^d} \leq 2, \quad N_1(\phi) \leq 240d.$$

Also $\|h_0\|_\infty \leq 1$ and $\|h'_0\|_\infty \leq \pi d < 4d$. The exact checker verifies, at $d = 128$ and by decreasing successive ratios thereafter,

$$60000d^2(49/90)^d < 1, \quad 60000d^2w^d < 1, \quad 610d2^{-d} < 1. \quad (9)$$

Thus $N_1(e) \leq d$, while (7) gives $N_1(C) \leq 610d$. It follows that $N_1(f_F - h_0) \leq 855d$.

For $t \leq 1/4$, integration from the nearest integer uses $f_F - h_0 = 0$ there and the monotonicity of h , giving the same N_0 bound. For $t \geq 1/4$, direct weighted amplitudes give

$$\frac{|f_F - h_0|}{h(t)^d} \leq 2 + 1 + \frac{1}{2} + 1 < 5,$$

using $|e|/h(t)^d \leq tN_0(e) < 1/2$. Division by $t \geq 1/4$ costs at most four. Hence $N_0(f_F - h_0) \leq 855d$ and $\|f_F - h_0\|_d \leq 1710d$.

Now $G = \text{ev}(I - QV_a|_X)^{-1}(f_F - h_0)$, so (5) proves its bound. Equation (8) and $31000d^2w^d < 1$ give $|F| \leq 3L^d$. Therefore $|FG + F^2/2| \leq (12825d + 9/2)L^{2d} \leq 13000dL^{2d}$. $\square$

This estimate controls all Taylor orders. It does not extend the unregularized orbit formula beyond its convergence range.

5 A uniform saddle comparison

Put $t_R = \arctan R$. For even $3d \leq N \leq 33d/10$, equations (2) and (8) give

$$[t^N]P_m(t) = 2[t^N]\mathcal{G}(t)^d + E_F - E_2 + E_3 + E_{\cos}. \quad (10)$$

The factor two follows because the two nearest branches have equal even coefficients after composition. Since $|\tan t| \leq \tan |t|$, Cauchy's estimate on $|t| = t_R$ and Lemmas 2.1 and 4.1 give

$$\begin{aligned} |E_F| &\leq 31000d^2(R\beta)^d t_R^{-N}, \\ |E_2| &\leq 13000d(RL)^{2d} t_R^{-N} < 13000d(R\beta)^d t_R^{-N}, \\ |E_3| &< 220(\log_2 d + 4)^2 2^d. \end{aligned} \quad (11)$$

The last inequality holds since $33/10 < 7/2$. Also $|E_{\cos}| \leq Cd$ for an absolute constant C , by Cauchy's estimate for $\log \cos t$ on $|t| = 1 < \pi/2$.

For $3 \leq \sigma \leq 33/10$, let t_σ be the positive saddle and set

$$\mu(t) = \frac{t\mathcal{G}'(t)}{\mathcal{G}(t)}, \quad \mu(t_\sigma) = \sigma, \quad \Phi(\sigma) = \log \mathcal{G}(t_\sigma) - \sigma \log t_\sigma.$$

The logarithmic derivative is the mean of the positive coefficient law of $\mathcal{G}$, and its derivative in $\log t$ is the strictly positive variance. Thus the saddle is unique.

Lemma 5.1. *There are constants $c_0 > 0$ and d_0 such that, for every $d \geq d_0$ and every integer N with $3 \leq N/d \leq 33/10$,*

$$[t^N]\mathcal{G}(t)^d \geq c_0 d^{-1/2} e^{d\Phi(N/d)}.$$

Moreover,

$$\Phi(\sigma) - \{\log(R\beta) - \sigma \log t_R\} > \frac{1}{400} \quad (3 \leq \sigma \leq 33/10). \quad (12)$$

Proof. Normalize the coefficients of $\mathcal{G}(t_\sigma z)$ by $\mathcal{G}(t_\sigma)$ to obtain an integer-valued law. The saddle interval is compact in $(0, \pi/2)$. The means are σ , the variances have positive lower and finite upper bounds, and there are exponential moments on one common neighborhood. The probabilities at one and two are uniformly positive, giving a uniform span-one Fourier gap away from zero. The characteristic functions have a uniform third-order expansion near zero. Fourier inversion, split between a shrinking neighborhood of zero and its complement, gives the lattice local limit theorem uniformly in σ . The mass at the exact integer mean N of the d -fold law is therefore at least $c_0/\sqrt{d}$ for all sufficiently large d . This proves the coefficient bound.

For the rate inequality, the rational interval checker establishes

$$.7252640 < t_3 < .7252641, \quad .7752157 < t_{33/10} < .7752158.$$

It uses directed Taylor bounds for sine and cosine, Machin's formula for π , and 32 factors of B . The omitted contribution to the mean is at most $a2^{-31}$, and the product truncation is a lower bound. The 20-term alternating arctangent sum is a rigorous lower bound for t_R ; using it preserves the required lower ratio.

At $\sigma = 3$, the resulting lower bound for $\mathcal{G}(t_\sigma)t_\sigma^{-\sigma}/((R\beta)t_R^{-\sigma})$ exceeds $6/5$. At $\sigma = 33/10$, its tenth power exceeds $103/100$. The corresponding logarithmic margins are greater than $1/6$ and $3/1030$, respectively, by $\log(1+x) \geq x/(1+x)$; both exceed $1/400$. Finally $\Phi' = -\log t_\sigma$, so Φ is concave. Subtracting the affine error exponent preserves concavity, whose minimum on the interval is at an endpoint. This proves (12). $\square$

Proof of Theorem 1.1. The first two errors in (11), divided by the positive term in Lemma 5.1, are bounded by a polynomial in d times $e^{-d/400}$, uniformly in N/d . Also

$$e^{\Phi(\sigma)} > (R\beta)t_R^{-\sigma} > \beta/R^2 > 3.$$

Thus the positive term exponentially dominates E_3 and $E_{\cos}$ as well. Equation (10) is strictly positive for one common sufficiently large d throughout $3d \leq N \leq 33d/10$. The established all-order result [2] covers $d < N < 3d$ and completes the proof. $\square$

6 Verification and remaining questions

The included standard-library checker proves every new scalar side condition using exact rational or outward integer interval arithmetic. It records the two saddle brackets and endpoint ratios. The existence threshold from the uniform local limit theorem has not been made effective. Consequently no finite table is used to fill an unspecified complement.

A separate model comparison suggests a sharper crossing. If t_k solves $\mu(t_k) = \sigma/k$, define $\Phi_k = k \log \mathcal{G}(t_k) - \sigma \log t_k$. A second rational checker certifies that the unique crossing $\Phi_1 = \Phi_2$ has slope 2σ between 6.662966 and 6.662968, strictly below $20/3$. Uniqueness follows from $(\Phi_2 - \Phi_1)' = \log(t_1/t_2) > 0$. This is a statement about the positive model only. Identifying it with $\lim N_m/m$, or proving that this limit exists, requires a sign-sensitive comparison of the actual first and second Schur terms. Neither is claimed here.

Editorial note (ProveIt, 2026-10-01). Both are proved in `../Thue_Morse_Integer_Pressure_First_Negative/`: $\lim N_m/m$ exists and equals the slope $\gamma = 2\sigma$ of this crossing, with $N_m = \gamma m - \Lambda^{-1} \log \log(2m) + O(1)$, where $\Lambda = \log(t_1/t_2) > 0$ at the crossing. The last sentence printed by `checks/verify_model_crossing.py` (“Actual-pressure identification remains unproved”) is superseded in the same way.

Editorial note (ProveIt, 2026-10-01). This note is the tenth of thirteen packages written in one series and filed together on 2026-10-01 beside the repository manuscript `../Thue_Morse_Integer_Pressure/` in the Fabius drafts tree (batch 72 of the repository’s `docs/incoming/`; unreviewed). In logical order they are the positivity series `../Thue_Morse_Integer_Pressure_First_Positive/`, `../Thue_Morse_Integer_Pressure_Higher_Positive/`, `../Thue_Morse_Integer_Pressure_Positive_Triangle/`, `../Thue_Morse_Integer_Pressure_Feedback_Boundary/`, `../Thue_Morse_Integer_Pressure_Beyond_Boundary/`, `../Thue_Morse_Integer_Pressure_Linear_Region/` and `../Thue_Morse_Integer_Pressure_Full_Range/`, and the sign series, which imports the full-range theorem: `../Thue_Morse_Integer_Pressure_Sign_Changes/`, `../Thue_Morse_Integer_Pressure_Sign_Densities/`, `../Thue_Morse_Integer_Pressure_Negative_Bound/`, `../Thue_Morse_Integer_Pressure_First_Negative/` and `../Thue_Morse_Integer_Pressure_Cluster_Asymptotics/`, with the dataset `../Thue_Morse_Integer_Pressure_First_Negative_Data/` (no article). An earlier version of `../Thue_Morse_Integer_Pressure_Full_Range/`, *The Full Pressure Positivity Range for Large Integer Orders* (orders $m \geq 4096$), delivered in the same batch, is superseded by it and was not filed. Here [1], cited as *Thue–Morse Integer Pressure*, is `../Thue_Morse_Integer_Pressure/`, whose title is *Integer Pressure and a Missing Taylor Coefficient* (the version cited differs from the filed manuscript only by dated editorial changes, none of them mathematical); [2] is `../Thue_Morse_Integer_Pressure_Full_Range/`; and [3] is `../Thue_Morse_Integer_Pressure_Sign_Changes/`. The copies bundled under `inputs/` were not filed.

References

- [1] V. Reshetnikov, *Thue–Morse Integer Pressure*, repository research manuscript, pinned at commit 63a7a325109ba611a1816b61dfd0eb072b896a7a. Source included in the package.
https://github.com/VladimirReshetnikov/ProveIt/blob/63a7a325109ba611a1816b61dfd0eb072b896a7a/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Integer_Pressure/article.tex
- [2] *The Full Pressure Positivity Range at Every Integer Order*, research report prepared for V. Reshetnikov with OpenAI, 1 October 2026. Unchanged PDF, TeX and source archive included. Sections 2, 4 and 5 contain the tangent, Green and weighted-residual inputs.
- [3] *Infinite Taylor Sign Changes in Integer Thue–Morse Pressure*, research report prepared for V. Reshetnikov with OpenAI, 1 October 2026. Unchanged PDF included; it supplies existence of the first negative degree.